Supplementary Material

Beyond model complexity: a pre-registered evaluation of event-level alarm protocols for spacecraft telemetry anomaly detection

This file accompanies the manuscript. Every number is inserted from the study's evidence file (thesis/evidence/evidence.json), which records the result file each value comes from. Section S4 reports the validation analyses of the second frozen plan (POSTHOC_PLAN_V2.md, SHA-256 prefix 217648e6805f, frozen 2026-10-25T12:14:37). Abbreviations: A0–A3, study arms; BH, Benjamini–Hochberg; ACI, adaptive conformal inference; M1, M2, Missions 1 and 2; FA, false alarms.

S1. Pre-registration, deviation log and validation plan

The following documents are provided as separate files in the repository and in the archived release:

- research_design/preregistration/PREREG_TELEMETRY_V1.md: the pre-registration, frozen as 09a98f5e160e, with 28 files hashed;
- research_design/preregistration/DEVIATIONS.md: every departure from the pre-registration, with its timing;
- research_design/preregistration/freeze_09a98f5e160e.json and freeze.json: the original freeze and the re-issued one (681cc7ddec86);
- research_design/preregistration/POSTHOC_PLAN_V2.md and posthoc_v2_freeze.json: the validation plan (P1–P8), frozen before any of its analyses was run.

S2. Results reported on ESA-ADB

Table S1. Results reported on ESA-ADB. *Primary scores as published; the protocols differ, so the numbers are not comparable.*

Study	Protocol	Method	Reported score
Kotowski et al. (2026)	official split; full and lightweight channels	TimeEval baselines, detector-specific thresholds	M1 full 0.061; M2 full 0.079; M1 lightweight 0.786
Şahin (2026)	official code; 24 / 19 channels; 16 seeds; pre-registered	telecommand conditioning; split, sliding and adaptive conformal thresholds	conditioning negative; point-level ACI collapses the event score
Nguyen et al. (2026)	own preprocessing; uncorrected event-wise metrics	supervised CNN and graph networks against unsupervised detectors	not comparable
Brancato et al. (2027)	Mission 1, hierarchical protocol (as stated in the abstract)	similarity search (SMED); quantile GP regression with retraining	M1 full 0.902 (SMED)
Alcarria et al. (2026)	monthly incremental retraining	LSTM autoencoder; PCA, k-means and LOF; adaptive thresholds	M1 0.700; M2 0.698
Allegrini and Pompei (2026)	Kaggle split of Mission 1, hidden test	supervised hierarchical ensemble	0.853 (private leaderboard)
Heydari et al. (2026)	two missions, 77 shared channels	threshold transfer under alert budgets	policy diagnostics
This study	official split; full target channels; pre-registered	AR(8), XGBoost and TCN under one event-level layer	M1 0.428 (AR(8)), 0.544 (trees)

Note: GP, Gaussian process; LOF, local outlier factor; SMED, the similarity-search detector of Brancato et al.

S3. Models and pre-registered arms

Table S2. Model settings. *No setting was tuned on test data; the supervised scorers received no hyperparameter search.*

Model	Setting	Value
AR(8)	order; forecast horizon; maximum fitted rows	8; 16 steps; 500,000
AR(8)	out-of-fold blocks; purge	5 contiguous blocks; 1 d on either side
AR(8)	score	absolute forecast error / max(residual scale, 0.05)
XGBoost (A2)	trees; depth; learning rate	400; 8; 0.1
XGBoost (A2)	row and column subsampling; minimum child weight	0.8; 0.8; 5
XGBoost (A2)	training rows; classes; class weights	2,000,000 (half labelled); 3; [1.0, 5.0, 5.0]
TCN (A3)	levels; kernel; width; receptive field	8; 3; 64; 1021 steps
TCN (A3)	cross-channel mixing; parameters (M1)	every 2 levels; 220,739
TCN (A3)	optimiser; steps; learning rate; weight decay; dropout	AdamW, one-cycle; 1,500; 0.001; 0.0001; 0.1
TCN (A3)	batch; window; share of windows with a labelled sample	8; 2,048 steps; 0.5
A2, A3	per-step features	asinh z; log(1 + s); its EWMA over 1 h, 12 h, 3 d; deviation from trailing 1 d mean; forecast indicator

Table S3. Secondary metrics, Mission 1. *Conformal-BH layer, test half, seed means.*

Arm	Precision	Recall	Alarming precision	ADTQC	Affiliation	Rare-only runs	Channel	Subsystem
A0	0.061	0.462	0.276	0.821	0.564	81	0.219	0.308
A1	0.667	0.176	0.139	0.862	0.445	11	0.077	0.134
A2	0.627	0.415	0.213	0.732	0.627	136	0.248	0.348
A3	0.211	0.266	0.326	0.646	0.463	42	0.106	0.218

Note: ADTQC, alarm detection timing quality; affiliation, affiliation-based $F_{0.5}$; rare-only runs, alarm runs touching only rare events; channel and subsystem, channel- and subsystem-aware $F_{0.5}$.

Table S4. Secondary metrics, Mission 2. *Conformal-BH layer, test half, seed means.*

Arm	Precision	Recall	Alarming precision	ADTQC	Affiliation	Rare-only runs
A0	0.982	0.892	0.895	0.897	0.931	351
A1	1.000	0.580	0.857	0.930	0.733	238
A2	0.975	0.710	0.901	0.969	0.814	277
A3	0.958	0.676	0.885	0.779	0.791	268

Note: ADTQC, alarm detection timing quality; affiliation, affiliation-based $F_{0.5}$; rare-only runs, alarm runs touching only rare events.

Table S5. Alarm layers on identical scores. *Test-half event-wise $F_{0.5}$, seed means.*

Mission	Arm	Threshold	Conformal-BH	Rolling BH	ACI	ACI + events
1	A1	0.428	0.428	0.101	0.0001	0.008
1	A2	0.613	0.544	0.087	0.0000	0.046
1	A3	0.282	0.205	0.323	0.0001	0.003
1	A3, no residual inputs	0.276	0.249	0.000	0.0001	0.005
1	A3, two classes	0.363	0.196	0.084	0.0001	0.000
1	A1, lightweight	0.350	0.025	0.130	0.0001	0.000
2	A1	0.977	0.874	0.356	0.0007	0.000
2	A2	0.819	0.904	0.244	0.0007	0.013
2	A3	0.853	0.875	0.487	0.0014	0.005
2	A1, lightweight	0.978	0.978	0.529	0.0017	0.000

Note: ACI and ACI + events are the point-level comparator as pre-registered (γ tuned to a point-wise rate).

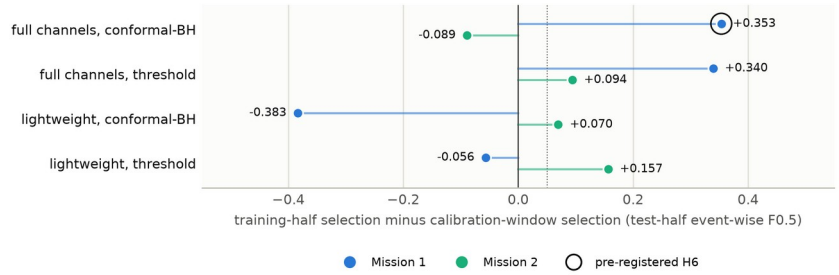


Fig. S1. Generality of the training-half protocol's effect. *Training-half protocol minus calibration-window protocol for AR(8), test-half event-wise $F_{0.5}$, by mission, channel set and alarm layer. The ringed point is the pre-registered H6 comparison; all other points are exploratory single runs.*

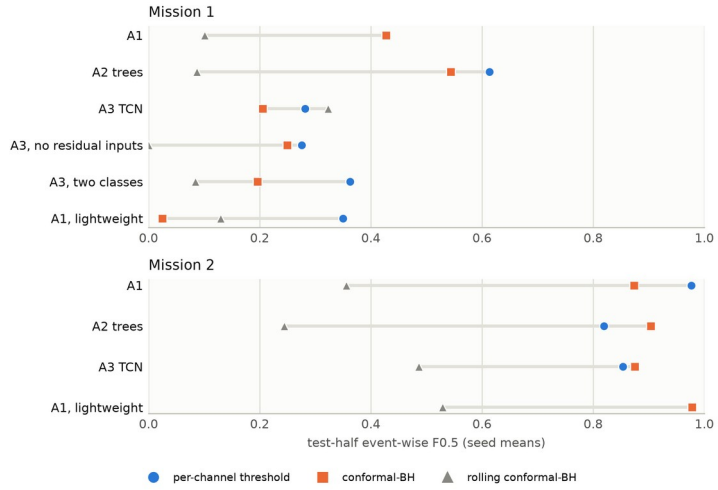


Fig. S2. Alarm layers on identical scores. *Test-half event-wise $F_{0.5}$ of the same scores under the threshold, conformal-BH and rolling layers (seed means). Coinciding markers mean equal values; ACI is omitted because all its values are below 0.002.*

Table S6. Per-seed results of the supervised arms. *Mission 1, conformal-BH layer, test half (pre-registered seeds 0–4).*

Seed	A2 $F_{0.5}$	A2 FA / yr	A3 $F_{0.5}$	A3 FA / yr
0	0.653	1.7	0.085	67.7
1	0.157	33.9	0.358	4.1
2	0.504	4.6	0.135	9.7
3	0.635	2.0	0.158	13.7
4	0.769	0.1	0.290	9.1

Table S7. Exploratory phase. *AR(8) with the layer selected on the calibration window, and the label-free baselines; test-half $F_{0.5}$ (before the pre-registration).*

Detector	M1 full	M1 lightweight	M2 full	M2 lightweight
AR(8), conformal-BH	0.0744	0.408	0.962	0.908
AR(8), threshold	0.0882	0.407	0.882	0.821
out-of-limit rule	0.0007	0.202	0.894	0.965
GlobalSTD5	0.0029	0.253	0.063	0.075

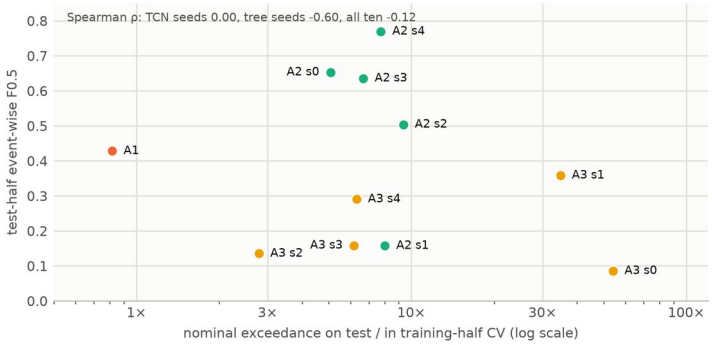


Fig. S3. Calibration shift against test score (exploratory). *For each supervised run on Mission 1 (BH layer), the ratio of the nominal test exceedance rate to the cross-validation rate at the* $1 - 10^{-4}$ *quantile, against test-half event-wise* $F_{0.5}$.

S4. Validation analyses (P1–P8)

All analyses in this section were specified and frozen before they were run, after the confirmatory analysis. They use test halves that had been used before.

P3, exact bootstrap. The block-decomposable metric of Appendix A was compared with the official value on 302 stored arm–metric pairs; the largest absolute difference was 2.1e-16 (criterion passed).

Table S8. P1 decomposition cells, Mission 1, target channels, conformal-BH layer. *Each cell is a full protocol scored once on the test half. Reproduction of A0 and A1: passed.*

Cell	Scores	Pool	Selection data	Grid	$F_{0.5}$	FA / yr	Level	k
c0	S0	cal	cal	G0	0.074	91.6	0.01	10
c1	S0	cal	cal	G1	0.074	91.6	0.01	10
c2	S½	cal	cal	G1	0.084	52.0	0.001	10
c3	S1	cal	cal	G1	0.084	52.0	0.001	10
c4	S1	half	cal	G1	0.127	34.6	0.01	10
c5	S1	half	half	G1	0.428	1.1	0.0001	10
r1	S1	half	half	G0	0.493	1.6	0.0003	10
r2	S1	half	cal	G0	0.127	34.6	0.01	10
r3	S1	cal	cal	G0	0.084	52.0	0.001	10
r4	S½	cal	cal	G0	0.084	52.0	0.001	10

Note: Forward path c0 → c5 changes G, F, σ, P, D in turn; reverse path c5 → r1 → r2 → r3 → r4 → c0 changes G, D, P, σ, F. S½ is A1's forecast errors on A0's residual scale.

Table S9. P1 steps, Mission 1, target channels, conformal-BH layer. *Change in test-half* $F_{0.5}$ *when one factor is switched, with exact block-bootstrap 95 % intervals, and the verdict of the pre-set rule.*

Factor	Forward path	Reverse path	Verdict
grid	+0.000 [+0.000, +0.000]	-0.065 [-0.167, +0.014]	order-dependent
AR(8) fit data	+0.010 [-0.009, +0.035]	+0.010 [-0.009, +0.033]	not material
residual scale	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	not material
calibration pool	+0.043 [+0.015, +0.083]	+0.043 [+0.014, +0.081]	not material
selection data	+0.301 [+0.156, +0.436]	+0.366 [+0.234, +0.500]	material

Note: The rule defines *material* for gains only (≥ +0.05 on both paths); a factor that lowers the score on both paths is *not material* under it.

Table S10. P1 decomposition cells, Mission 1, target channels, threshold layer. *Each cell is a full protocol scored once on the test half. Reproduction of A0 and A1: passed.*

Cell	Scores	Pool	Selection data	Grid	$F_{0.5}$	FA / yr	Level	k
c0	S0	cal	cal	G0	0.088	64.4	0.0001	10
c1	S0	cal	cal	G1	0.088	64.4	0.0001	10
c2	S½	cal	cal	G1	0.085	49.3	1e-05	10
c3	S1	cal	cal	G1	0.085	49.3	1e-05	10
c4	S1	half	cal	G1	0.231	13.1	0.0001	10
c5	S1	half	half	G1	0.428	1.1	0	10
r1	S1	half	half	G0	0.428	1.1	0	10
r2	S1	half	cal	G0	0.231	13.1	0.0001	10
r3	S1	cal	cal	G0	0.085	49.3	1e-05	10
r4	S½	cal	cal	G0	0.085	49.3	1e-05	10

Note: Forward path c0 → c5 changes G, F, σ, P, D in turn; reverse path c5 → r1 → r2 → r3 → r4 → c0 changes G, D, P, σ, F. S½ is A1's forecast errors on A0's residual scale.

Table S11. P1 steps, Mission 1, target channels, threshold layer. *Change in test-half $F_{0.5}$ when one factor is switched, with exact block-bootstrap 95 % intervals, and the verdict of the pre-set rule.*

Factor	Forward path	Reverse path	Verdict
grid	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	not material
AR(8) fit data	-0.003 [-0.021, +0.017]	-0.003 [-0.020, +0.016]	not material
residual scale	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	not material
calibration pool	+0.145 [+0.071, +0.263]	+0.145 [+0.070, +0.265]	material
selection data	+0.197 [+0.060, +0.327]	+0.197 [+0.063, +0.338]	material

Note: The rule defines *material* for gains only (≥ +0.05 on both paths); a factor that lowers the score on both paths is *not material* under it.

Table S12. P1 decomposition cells, Mission 1, lightweight channels, conformal-BH layer. *Each cell is a full protocol scored once on the test half. Reproduction of A0 and A1: passed.*

Cell	Scores	Pool	Selection data	Grid	$F_{0.5}$	FA / yr	Level	k
c0	S0	cal	cal	G0	0.408	5.0	0.001	3
c1	S0	cal	cal	G1	0.408	5.0	0.001	3
c2	S½	cal	cal	G1	0.340	3.0	0.01	10
c3	S1	cal	cal	G1	0.340	3.0	0.01	10
c4	S1	half	cal	G1	0.191	20.6	0.01	3
c5	S1	half	half	G1	0.025	11.9	0.01	30
r1	S1	half	half	G0	0.025	11.9	0.01	30
r2	S1	half	cal	G0	0.191	20.6	0.01	3
r3	S1	cal	cal	G0	0.413	3.1	0.0003	3
r4	S½	cal	cal	G0	0.413	3.1	0.0003	3

Note: Forward path c0 → c5 changes G, F, σ, P, D in turn; reverse path c5 → r1 → r2 → r3 → r4 → c0 changes G, D, P, σ, F. S½ is A1's forecast errors on A0's residual scale.

Table S13. P1 steps, Mission 1, lightweight channels, conformal-BH layer. *Change in test-half $F_{0.5}$ when one factor is switched, with exact block-bootstrap 95 % intervals, and the verdict of the pre-set rule.*

Factor	Forward path	Reverse path	Verdict
grid	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	not material
AR(8) fit data	-0.068 [-0.159, +0.024]	+0.005 [-0.078, +0.083]	order-dependent
residual scale	+0.000 [+0.000, +0.000]	+0.000 [-0.017, +0.022]	not material
calibration pool	-0.150 [-0.258, -0.049]	-0.223 [-0.335, -0.110]	order-dependent
selection data	-0.166 [-0.243, -0.043]	-0.166 [-0.245, -0.042]	not material

Note: The rule defines *material* for gains only (≥ +0.05 on both paths); a factor that lowers the score on both paths is *not material* under it.

Table S14. P1 decomposition cells, Mission 2, target channels, conformal-BH layer. *Each cell is a full protocol scored once on the test half. Reproduction of A0 and A1: passed.*

Cell	Scores	Pool	Selection data	Grid	$F_{0.5}$	FA / yr	Level	k
c0	S0	cal	cal	G0	0.962	3.4	0.01	30
c1	S0	cal	cal	G1	0.962	3.4	0.01	30
c2	S½	cal	cal	G1	0.965	2.9	0.01	30
c3	S1	cal	cal	G1	0.966	2.9	0.01	30
c4	S1	half	cal	G1	0.962	1.1	0.01	30
c5	S1	half	half	G1	0.874	0.0	0.001	30
r1	S1	half	half	G0	0.874	0.0	0.001	30
r2	S1	half	cal	G0	0.962	1.1	0.01	30
r3	S1	cal	cal	G0	0.966	2.9	0.01	30
r4	S½	cal	cal	G0	0.965	2.9	0.01	30

Note: Forward path c0 → c5 changes G, F, σ, P, D in turn; reverse path c5 → r1 → r2 → r3 → r4 → c0 changes G, D, P, σ, F. S½ is A1's forecast errors on A0's residual scale.

Table S15. P1 steps, Mission 2, target channels, conformal-BH layer. *Change in test-half $F_{0.5}$ when one factor is switched, with exact block-bootstrap 95 % intervals, and the verdict of the pre-set rule.*

Factor	Forward path	Reverse path	Verdict
grid	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	not material
AR(8) fit data	+0.003 [-0.004, +0.011]	+0.003 [-0.004, +0.011]	not material
residual scale	+0.001 [-0.000, +0.002]	+0.001 [-0.000, +0.002]	not material
calibration pool	-0.004 [-0.017, +0.015]	-0.004 [-0.018, +0.014]	not material
selection data	-0.088 [-0.134, -0.054]	-0.088 [-0.134, -0.055]	not material

Note: The rule defines *material* for gains only (≥ +0.05 on both paths); a factor that lowers the score on both paths is *not material* under it.

Table S16. P1 decomposition cells, Mission 2, target channels, threshold layer. *Each cell is a full protocol scored once on the test half. Reproduction of A0 and A1: passed.*

Cell	Scores	Pool	Selection data	Grid	$F_{0.5}$	FA / yr	Level	k
c0	S0	cal	cal	G0	0.882	0.0	0	30
c1	S0	cal	cal	G1	0.882	0.0	0	30
c2	S½	cal	cal	G1	0.880	0.0	0	30
c3	S1	cal	cal	G1	0.884	0.0	0	30
c4	S1	half	cal	G1	0.977	1.7	0.001	30
c5	S1	half	half	G1	0.977	1.7	0.001	30
r1	S1	half	half	G0	0.977	1.7	0.001	30
r2	S1	half	cal	G0	0.977	1.7	0.001	30
r3	S1	cal	cal	G0	0.884	0.0	0	30
r4	S½	cal	cal	G0	0.880	0.0	0	30

Note: Forward path c0 → c5 changes G, F, σ, P, D in turn; reverse path c5 → r1 → r2 → r3 → r4 → c0 changes G, D, P, σ, F. S½ is A1's forecast errors on A0's residual scale.

Table S17. P1 steps, Mission 2, target channels, threshold layer. *Change in test-half $F_{0.5}$ when one factor is switched, with exact block-bootstrap 95 % intervals, and the verdict of the pre-set rule.*

Factor	Forward path	Reverse path	Verdict
grid	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	not material
AR(8) fit data	-0.002 [-0.011, +0.005]	-0.002 [-0.011, +0.005]	not material
residual scale	+0.005 [-0.001, +0.013]	+0.005 [-0.001, +0.012]	not material
calibration pool	+0.092 [+0.054, +0.142]	+0.092 [+0.055, +0.143]	material
selection data	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	not material

Note: The rule defines *material* for gains only (≥ +0.05 on both paths); a factor that lowers the score on both paths is *not material* under it.

Table S18. P1 decomposition cells, Mission 2, lightweight channels, conformal-BH layer. *Each cell is a full protocol scored once on the test half. Reproduction of A0 and A1: passed.*

Cell	Scores	Pool	Selection data	Grid	$F_{0.5}$	FA / yr	Level	k
c0	S0	cal	cal	G0	0.908	9.2	0.01	30
c1	S0	cal	cal	G1	0.908	9.2	0.01	30
c2	S½	cal	cal	G1	0.913	18.3	0.001	10
c3	S1	cal	cal	G1	0.903	21.2	0.001	10
c4	S1	half	cal	G1	0.945	0.0	0.01	30
c5	S1	half	half	G1	0.978	0.6	0.001	10
r1	S1	half	half	G0	0.978	0.6	0.001	10
r2	S1	half	cal	G0	0.945	0.0	0.01	30
r3	S1	cal	cal	G0	0.903	21.2	0.001	10
r4	S½	cal	cal	G0	0.913	18.3	0.001	10

Note: Forward path c0 → c5 changes G, F, σ, P, D in turn; reverse path c5 → r1 → r2 → r3 → r4 → c0 changes G, D, P, σ, F. S½ is A1's forecast errors on A0's residual scale.

Table S19. P1 steps, Mission 2, lightweight channels, conformal-BH layer. *Change in test-half $F_{0.5}$ when one factor is switched, with exact block-bootstrap 95 % intervals, and the verdict of the pre-set rule.*

Factor	Forward path	Reverse path	Verdict
grid	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	not material
AR(8) fit data	+0.005 [-0.027, +0.035]	+0.005 [-0.027, +0.035]	not material
residual scale	-0.009 [-0.019, -0.002]	-0.009 [-0.019, -0.002]	not material
calibration pool	+0.042 [-0.022, +0.132]	+0.042 [-0.025, +0.135]	not material
selection data	+0.033 [+0.018, +0.052]	+0.033 [+0.018, +0.051]	not material

Note: The rule defines *material* for gains only (≥ +0.05 on both paths); a factor that lowers the score on both paths is *not material* under it.

Table S20. P2 forward chaining, Mission 1. *Per fold: events of the fold, $F_{0.5}$ and false alarms per year for the calibration-window protocol W and the training-history protocol H.*

Layer	Fold start	Events	W $F_{0.5}$	H $F_{0.5}$	H − W	W FA / yr	H FA / yr
conformal-BH	2003-01-01	9	0.092	0.238	+0.146	23.0	2.0
conformal-BH	2004-01-01	10	0.013	0.000	-0.013	271.4	0.0
conformal-BH	2005-01-01	10	0.008	0.833	+0.825	1172.8	0.0
conformal-BH	2006-01-01	10	0.357	0.357	+0.000	0.0	0.0
threshold	2003-01-01	9	0.000	0.345	+0.345	0.0	3.0
threshold	2004-01-01	10	0.016	0.000	-0.016	76.8	0.0
threshold	2005-01-01	10	0.682	0.769	+0.087	0.0	0.0
threshold	2006-01-01	10	0.357	0.278	-0.079	0.0	1.0

Note: Pooled over folds (BH): H − W = +0.455 [+0.189, +0.659]. Replication criterion met: no.

Table S21. P2 forward chaining, Mission 2. *Per fold: events of the fold, $F_{0.5}$ and false alarms per year for the calibration-window protocol W and the training-history protocol H.*

Layer	Fold start	Events	W $F_{0.5}$	H $F_{0.5}$	H − W	W FA / yr	H FA / yr
conformal-BH	2001-01-01	29	0.917	0.929	+0.012	0.0	0.0
conformal-BH	2001-04-01	29	0.957	0.977	+0.020	4.0	0.0
conformal-BH	2001-07-01	26	0.941	0.960	+0.020	7.9	4.0
threshold	2001-01-01	29	0.917	0.940	+0.023	0.0	0.0
threshold	2001-04-01	29	1.000	0.985	-0.015	0.0	0.0
threshold	2001-07-01	26	0.960	0.951	-0.009	4.0	4.0

Note: Pooled over folds (BH): H − W = +0.017 [-0.007, +0.044]. Replication criterion met: no.

Table S22. P4, ACI tuned on the event objective. *Chosen (α, γ, k) and test-half $F_{0.5}$ for the best setting overall, the best with $\gamma > 0$, and $\gamma = 0$ (a fixed per-channel conformal threshold).*

Arm	Best: α, γ, k	$F_{0.5}$	Best with $\gamma > 0$	$F_{0.5}$	$\gamma = 0$ $F_{0.5}$
M1 A1	0.0001, 0.0001, 10	0.472	0.0001, 0.0001, 10	0.472	0.261
M1 A2 seed 0	0.0001, 0, 100	0.252	0.001, 0.0001, 100	0.407	0.252
M1 A2 seed 1	0.001, 0.0001, 100	0.535	0.001, 0.0001, 100	0.535	0.212
M1 A2 seed 2	0.0001, 0, 100	0.218	0.001, 0.0001, 100	0.513	0.218
M1 A2 seed 3	0.001, 0.0001, 100	0.453	0.001, 0.0001, 100	0.453	0.265
M1 A2 seed 4	0.0001, 0, 100	0.290	0.001, 0.0001, 100	0.510	0.290
M2 A1	0.001, 0, 30	0.977	0.01, 0.0001, 30	0.925	0.977

Table S23. P5, registered anomaly-score arm, Mission 1. *Anomaly-only test-half $F_{0.5}$ with the layer selected on the anomaly-only objective.*

Arm	Runs	Mean	sd	Min	Max	FA / yr
A1	1	0.336	0.000	0.336	0.336	1.1
A2	5	0.087	0.091	0.011	0.232	44.7
A3	5	0.097	0.072	0.034	0.211	18.5

Note: A1 is deterministic; A2 and A3 use their anomaly log-odds.

Table S24. P5, registered anomaly-score arm, Mission 2. *Anomaly-only test-half $F_{0.5}$ with the layer selected on the anomaly-only objective.*

Arm	Runs	Mean	sd	Min	Max	FA / yr
A1	1	0.000	0.000	0.000	0.000	0.0
A2	5	0.247	0.176	0.081	0.500	3.0
A3	5	0.005	0.010	0.000	0.023	6.6

Note: A1 is deterministic; A2 and A3 use their anomaly log-odds.

Table S25. P6, seed extension. *Mission 1, conformal-BH layer: seed statistics with all available seeds, and the comparisons with the exact bootstrap (unadjusted, exploratory).*

Arm or comparison	Seeds	Mean $F_{0.5}$ / difference	sd	Min / 95 % CI	Max / p
A2	10	0.488	0.195	0.157	0.769
A3	9	0.176	0.093	0.082	0.358
H4: A3 − A1		-0.252		[-0.402, -0.043]	p = 0.020
H5: A3 − A2		-0.313		[-0.457, -0.112]	p = 0.002
X1: A2 − A1		+0.061		[-0.138, +0.268]	p = 0.528

Table S26. P7, operational metrics. *Test halves, conformal-BH layer; seed means for A2 and A3.*

Mission	Arm	Detected / events	Anomalies	Rare events	Delay, median [IQR] (h)	Within 1 h	Within 24 h	Runs per detected event	FA per 1000 nominal h
1	A0	42.0 / 91	17.0	25.0	0.35 [0.13, 6.54]	0.67	0.86	2.0	11.749
1	A1	16.0 / 91	8.0	8.0	0.33 [0.17, 3.73]	0.62	0.88	1.0	0.147
1	A2	37.8 / 91	11.8	26.0	1.07 [0.40, 2.52]	0.47	0.95	1.4	1.085
1	A3	24.2 / 91	11.8	12.4	2.02 [0.16, 5.75]	0.43	0.92	1.6	2.680
2	A0	323.0 / 362	1.0	322.0	0.16 [0.15, 0.19]	0.97	1.00	1.0	0.411
2	A1	210.0 / 362	0.0	210.0	0.16 [0.15, 0.19]	0.95	1.00	1.0	0.000
2	A2	257.2 / 362	0.8	256.4	0.07 [0.04, 0.10]	0.98	1.00	1.0	0.425
2	A3	244.8 / 362	0.8	244.0	0.27 [0.24, 0.30]	0.97	1.00	1.0	0.918

Note: Delay is first-alarm time minus event start for detected events; IQR, interquartile range.

P8, provenance. Re-running the frozen selection for the A1 arms reproduced the saved choice and every grid value: Mission 1 passed (largest difference 0), Mission 2 passed (largest difference 0).

S5. Code, data and reproduction

The repository contains the frozen code of the pre-registered study, the validation scripts (scripts/posthoc_v2/), the evidence builder (thesis/tools/build_evidence.py) and the manuscript build (paper/journal_v2/tools/). The ESA Anomaly Dataset is obtained from Zenodo (DOI 10.5281/zenodo.12528696). Derived score and alarm arrays will be deposited with the archived release.

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